

Autonomous Rover Navigation & Stereo Visual Odometry

1. Terrain Hazard Mapping

Navigational mast cameras execute stereo disparity matching at 10 Hz, generating 3D digital elevation meshes with 2-centimeter resolution up to 30 meters ahead.

2. Rocker-Bogie Chassis Control

Six independent hub-motor driven titanium wheels traverse boulders up to 45 centimeters in height with active wheel-walking torque vectoring on steep basalt sand dunes.

3. Path Planning Algorithmic Suite

Real-time Field D* pathfinding recalculates optimal traverse trajectories every 250 milliseconds, bypassing sand traps and maintaining maximum slip ratios below 18 percent.